

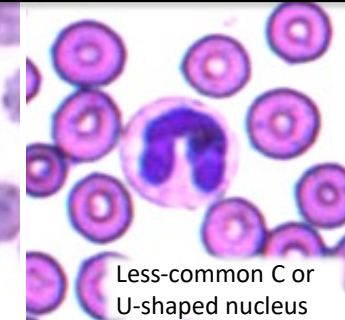
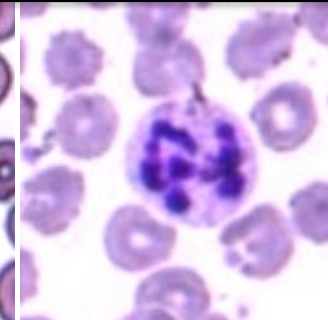
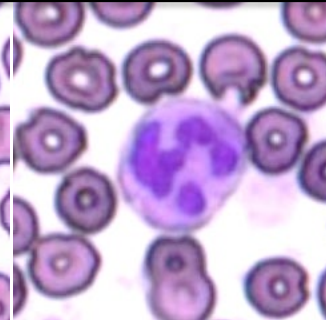
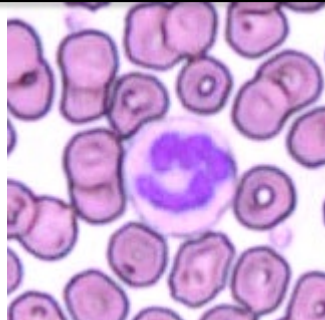
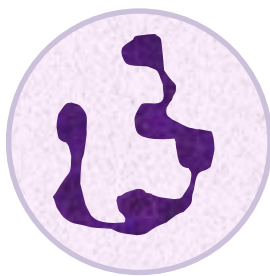
Neutrophil

Most common (40-60%)*

Cytoplasm – Light pinkish with small granules, occasional vacuole.

Nucleus – Dark purple with multiple connected shapes.

2-3x larger than RBC



Less-common C or U-shaped nucleus

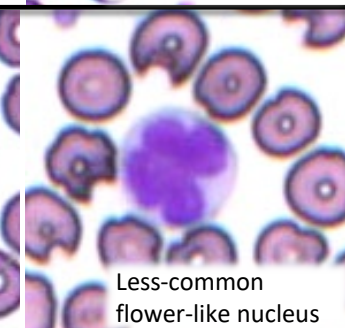
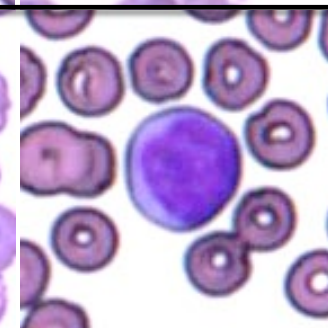
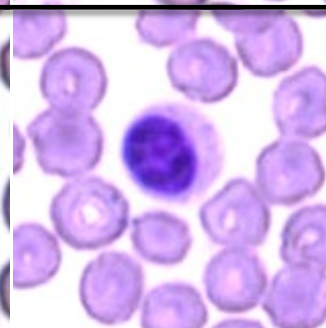
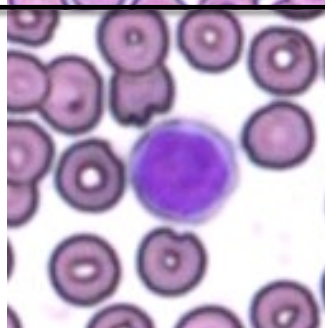
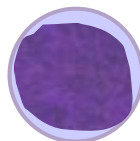
Lymphocyte

Common (20-40%)*

Cytoplasm – Light (typically bluish), clear with no granules.

Nucleus – Dark purple, rounded, fills up most of cell.

Nucleus slightly larger than RBC



Less-common flower-like nucleus

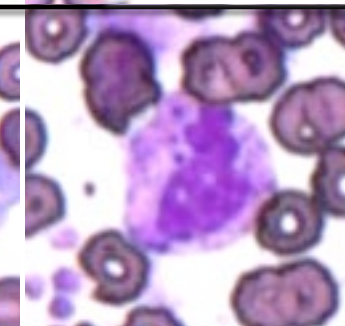
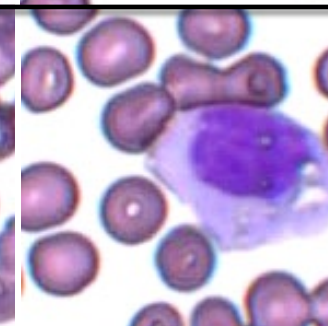
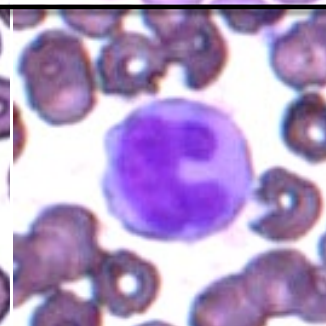
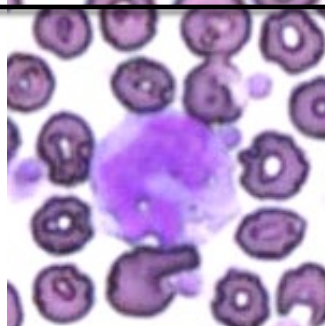
Monocyte

Less common (2-8%)*

Cytoplasm – Light (typically bluish), clear, no granules, some vacuoles.

Nucleus – Dark purple, irregularly-shaped, (Some C or Bean-shaped).

2-3x larger than RBC



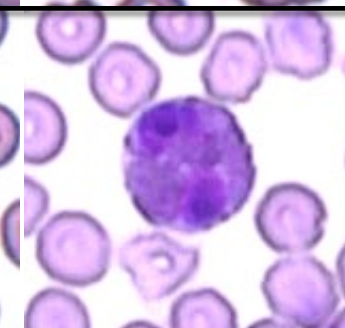
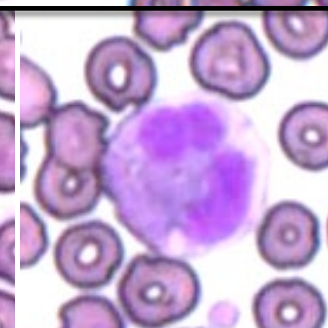
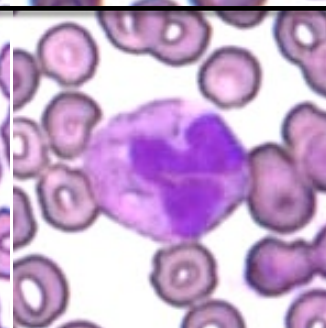
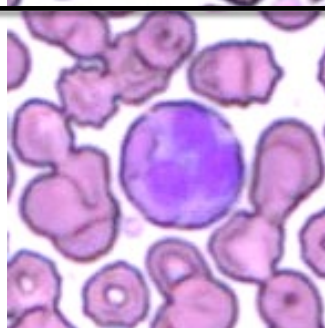
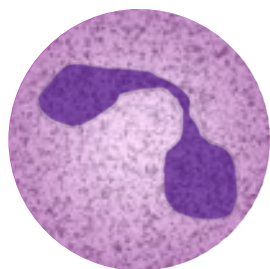
Eosinophil

Less common (1-4%)*

Cytoplasm – Darker red or purplish with medium-sized granules.

Nucleus – Dark purple with 2-3 round, connected shapes.

2-3x larger than RBC



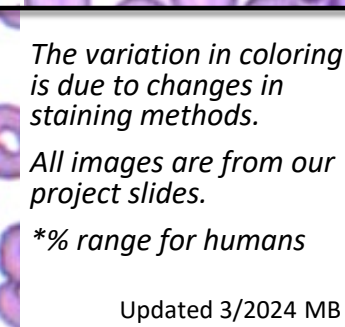
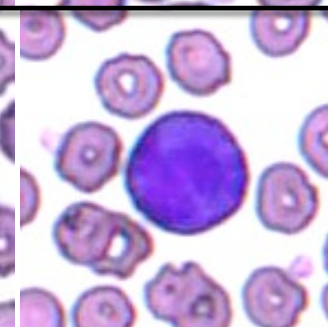
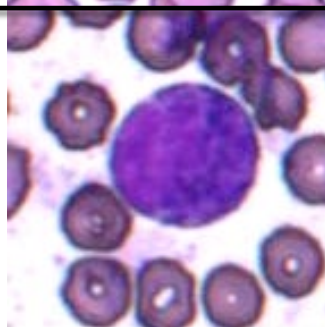
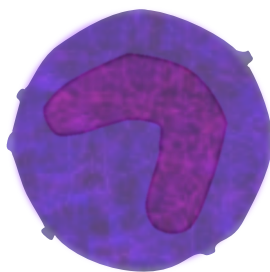
Basophil

Rare (<1%)*

Cytoplasm – Filled with large, dark purple granules.

Nucleus – Dark purple, bean-shaped, mostly hidden by granules.

2-3x larger than RBC



The variation in coloring is due to changes in staining methods.

All images are from our project slides.

*% range for humans

Neutrophil:

Most common (40-60%)*

Cytoplasm – Light pinkish with small granules, occasional vacuole.

Nucleus – Dark purple with multiple connected shapes.

2-3x larger than RBC

Platelet

Lymphocyte

Lymphocyte:

Common (20-40%)*

Cytoplasm – Light (typically bluish), clear with no granules.

Nucleus – Dark purple, rounded, fills up most of cell.

Nucleus slightly larger than RBC

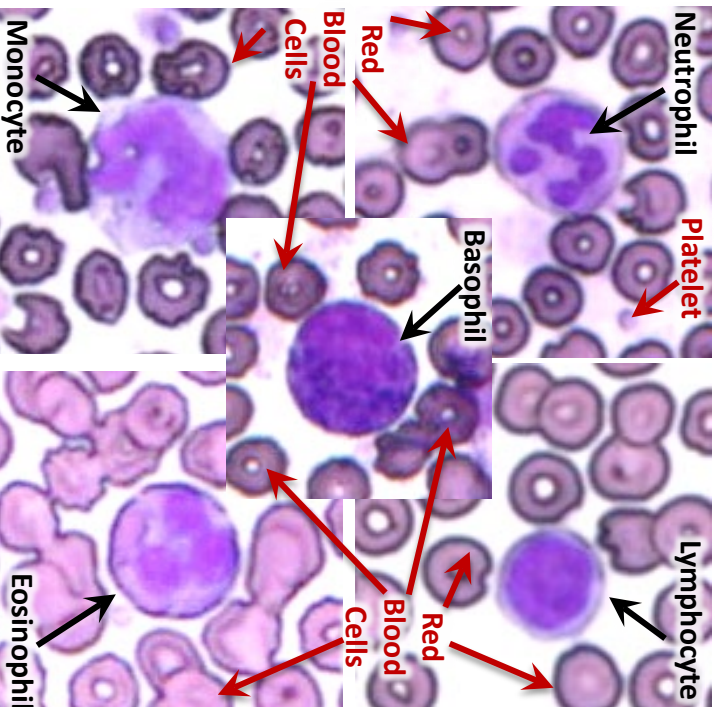
Monocyte:

Less common (2-8%)*

Cytoplasm – Light (typically bluish), clear, no granules, some vacuoles.

Nucleus – Dark purple, irregularly-shaped, (Some C or Bean-shaped).

2-3x larger than RBC



Basophil (center):

Rare (<1%)*

Cytoplasm – Filled with large, dark purple granules.

Nucleus – Dark purple, bean-shaped, mostly hidden by granules.

2-3x larger than RBC

**% range for humans*

Not White Blood Cells

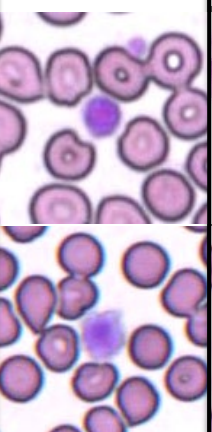
Burst Cells

Do not mark cells that have burst open, spilling their contents, unless they are still enough intact to clearly identify them.



Platelets

Some platelets are large and can be about the size of a red blood cell. They can sometimes be mistaken for lymphocytes, but they don't have the distinctive dark nucleus surrounded by lighter cytoplasm and a cell membrane..



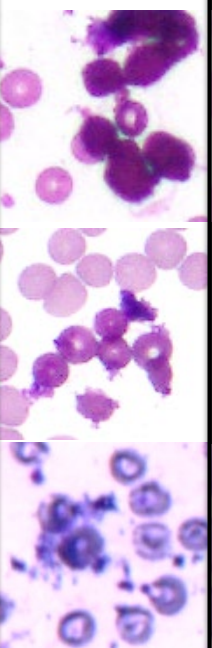
Cell Fakes

These are red blood cells with platelets or other debris on top – making them look like neutrophils or other white blood cells.



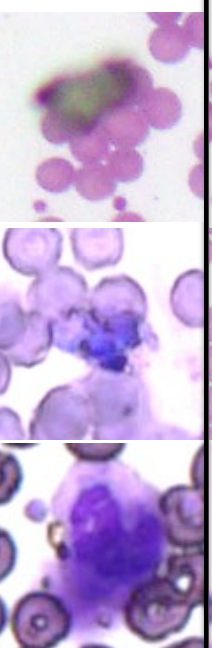
Strange Reds

Red blood cells that get overstained, have debris stuck to them, or otherwise look darker than normal.



General Debris

Do not mark cell-sized debris unless you are sure it is a white blood cell. Do not classify a white blood cell if debris is blocking too much of it to identify it.



The variation in coloring is due to changes in staining methods. All images are from our slides except for those with a reference.